Cheng Tang

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Research short recap

During my PhD, I spent time studying unsupervised learning methods, including the k-means method and principal component analysis, from a theoretical perspective. At Amazon, I applied large language models to product-inspired problems in natural language processing and information retrieval.

Gap years

Due to complicated situation, I was away from academia and the job market for several years. I took this opportunity to catch up on research topics in ML theory and LLM's. I am currently interested in research at the intersection of ML theory and applications, for example, domain adaptation, causal inference, and large language models.

Employment

Postdoc, (Incomplete for personal reasons)
Anonymous Institution, 2021

Applied Scientist II, New York, NY USA
AI Labs, Amazon Web Services, Jul. 2018 – Jan. 2021
Managers: Bing Xiang & Andrew Arnold

- Experience: I explored neural embedding models, transformer-based models (BERT and BART), and their applications to domain adaptation, reading comprehension, recommendation and ranking.

Research Internships

Research Intern, Cambridge, UK
Machine Intelligence group, Microsoft Research, Jun. 2017– Aug. 2017
Mentors: Alex Gaunt & Ryota Tomioka, Jul., 2017– Sep., 2017

- Project: Explored the application of a neural-logical model to the problem of understanding the hidden rules that govern the gene expression process.

Research Visitor, Tübingen, Germany
Theory of Machine Learning group, University of Tübingen, Germany, Mar. 2017– Jun. 2017
Mentor: Ulrike von Luxburg, Mar., 2017– Jun., 2017

- Project: Theoretical exploration of the role of sub-sampling in random forests

Education

The George Washington University, Washington, DC USA
Ph.D., Computer Science, May 2018

- Advisor: Claire Monteleoni

B.S., Mathematics, Magna Cum Laude, GPA 3.77/4.0, May 2012

Preprint and ongoing works

- C. Tang, D. Garreau, C. Monteleoni, "Multiple-source adaptation with the Hedge algorithm" (Preprint link: `https://tangch30.github.io/assets/files/msa_preprint.pdf`), 2026.
- C. Tang, Foundation of auto-differentiation and computational graphs, in-preparation.
- C. Tang, On the stability of hyperparameters during data scaling, in-preparation.
- C. Tang, "On the tightness of linear relaxation based robustness certification methods", 2022.
- C. Tang, Andrew Arnold, "Neural document expansion for ad-hoc information retrieval" (arXiv: `https://arxiv.org/pdf/2012.14005`), 2020.

Conference Proceedings

- C. Tang, "Exponentially convergent stochastic k-PCA without variance reduction", NeurIPS, 2019.
- C. Tang, D. Garreau, U. von Luxburg, "When do random forests fail?", Proceedings of 32nd Conference on Neural Information Processing Systems (NeurIPS), 2018.
- C. Tang and C. Monteleoni, "Convergence rate of stochastic k-means", AISTATS, 2017.
- C. Tang and C. Monteleoni, "On Lloyd's algorithm: new theoretical insights for clustering in practice," Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS), pp. 1280-1289, 2016.

Academic reviewing experience

AISTATS (2017, 2019, 2020, 2021), ICML (2015, 2018, 2019, 2020), NeurIPS (2016, 2017, 2018), ICLR (2017, 2019, 2020, 2021), AAAI (2019, 2020, 2021), TPAMI (2019, 2020)

Selected honors and awards

- Engineer Alumni Association Scholarship, SEAS, GWU, 2015– 2016
- Louis P. Wagman Endowment Fellowship, GWU, 2013
- Presidential Academic Scholarship, GWU, 2008– 2012
- Ranked 165/230,000, National College Entrance Exam (admitted to Fudan University), Sichuan province, China, 2008.

Academic Teaching Experience

- Discrete Structures II, Graduate Teaching Assistant, Department of Computer Science, GWU, Fall, 2016
- Machine Learning, Undergraduate Teaching Assistant, Department of Computer Science, GWU, Spring 2012
- Math and Politics, Undergraduate Teaching Assistant, Department of Mathematics, GWU, Fall, 2011

Computer Languages

Python (including deep-learning frameworks such as TensorFlow/PyTorch/MXNet), MATLAB, Unix/Linux